

LECTURE 3

LAST TIME

LECTURE 2: SUMMARY

Facts and data on derivatives markets

No arb

LoOP

CIP violations and regulation (“limits to arb”)

THIS TIME

LECTURE 2 OUTLINE

Hedging with forwards

The forward price

WARM-UP: PRICES VS. CASH FLOWS

Suppose you buy a portfolio with price \$5 at $t = 0$

What is your cash flow CF_0 ?

Suppose you buy a portfolio with price $-\$5$

What is your cash flow CF_0 ?

Do securities have negative prices in reality? Examples?

THE PRICE OF CHOCOLATE

CHOCOLATE AND COCOA

The price of chocolate is stable

The price of cocoa?

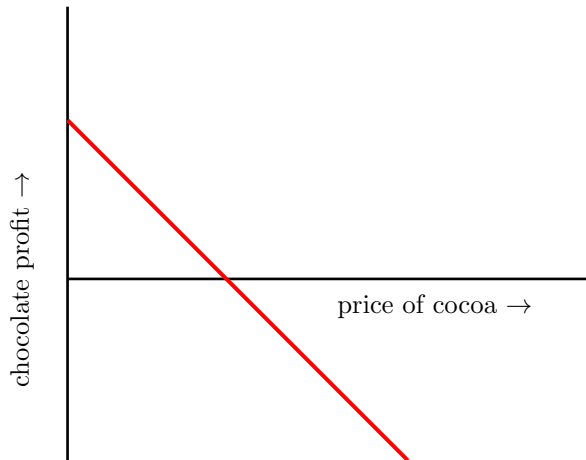
CHOCOLATE AND COCOA

The price of chocolate is stable

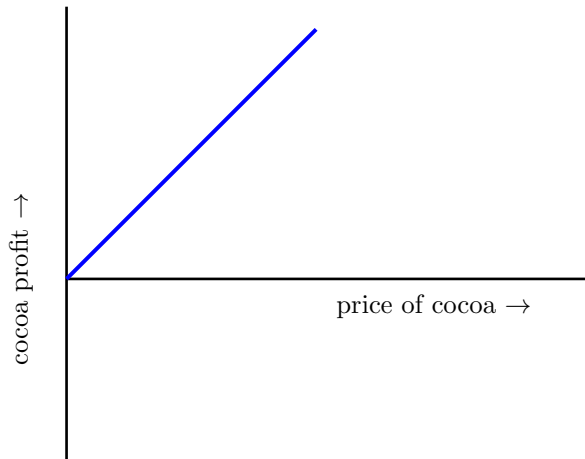
The price of cocoa?

How do chocolate companies avoid passing volatility on to you?

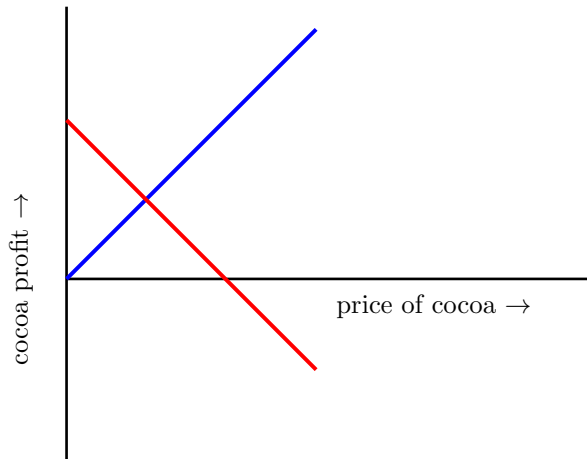
CHOCOLATE FIRM'S UNHEDGED EXPOSURE



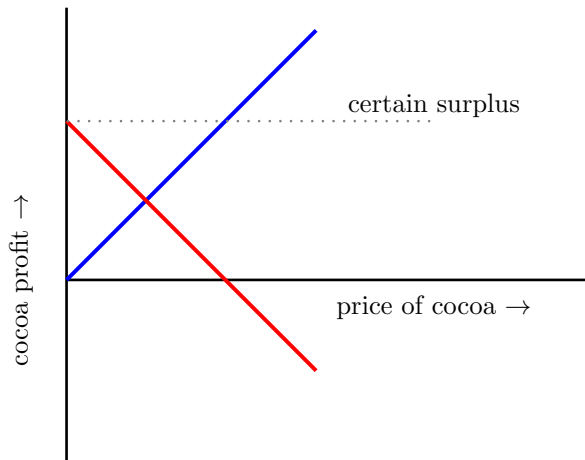
COCOA FIRM'S UNHEDGED EXPOSURE



DOUBLE COINCIDENCE OF WANTS



DOUBLE COINCIDENCE OF WANTS IN RISK



DIVISION OF SURPLUS

Cocoa and chocolate firms agree today on price to trade at tomorrow

No exchange today, no uncertainty about exchange tomorrow

$\implies$ both parties fully insured (to cocoa price risk)

Call this forward price $F_t(T)$

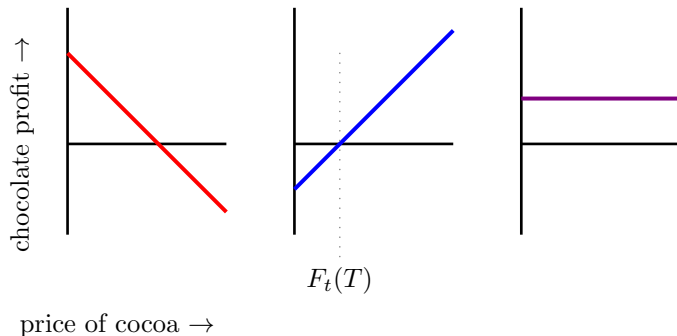
t is inception: when contract agreed

T is maturity: when delivery occurs

Chocolate firm long forward, cocoa firm short

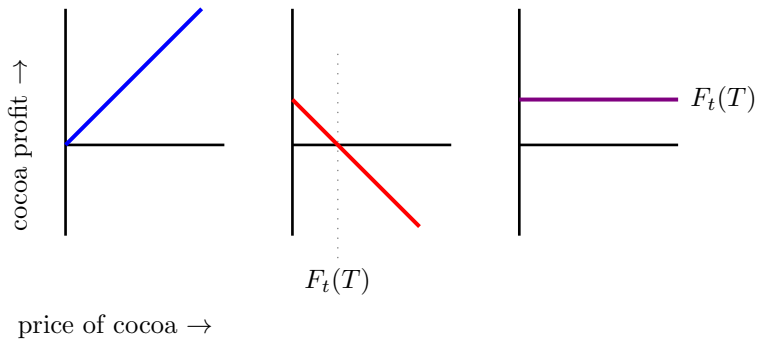
CHOCOLATE FIRM HEDGES

original exposure + long forward = hedged position



COCOA FIRM HEDGES

original exposure + short forward = hedged position



FORWARDS INSURE BOTH PARTIES

Forward contract allows both parties complete insurance

$\implies$ Will agree to forward contract

FORWARDS INSURE BOTH PARTIES

Forward contract allows both parties complete insurance

$\implies$ Will agree to forward contract. But at what price?

FORWARDS INSURE BOTH PARTIES

Forward contract allows both parties complete insurance

$\implies$ Will agree to forward contract. But at what price?

Reminders:

$F_t(T)$ is the price set at t for delivery at T)

What is the value of the forward today (at t)?

NOTE ON TERMINOLOGY

For now focus on world with just two dates $t = 0$ and $T = 1$

Form a portfolio just once, today

“Static” portfolio choice per Markowitz (hence CAPM)

Later: more dates $\implies$ portfolio at each one (“trading strategy”)

BACK TO CHOCOLATE

FORWARD PRICE OF COCOA

Firms agree to forward price of cocoa: $F_0(1)$. What is it?

Notation:

p_t is price of cocoa at $t = 0$ and $t = 1$ (p_1 uncertain)

r is the risk-free rate

X_t is price/payoff of the forward contract at t

FORWARD PRICE OF COCOA

Firms agree to forward price of cocoa: $F_0(1)$. What is it?

Notation:

p_t is price of cocoa at $t = 0$ and $t = 1$ (p_1 uncertain)

r is the risk-free rate

X_t is price/payoff of the forward contract at t

Question: What is difference between X and F ?

FORWARD PRICE OF COCOA

Firms agree to forward price of cocoa: $F_0(1)$. What is it?

Notation:

p_t is price of cocoa at $t = 0$ and $t = 1$ (p_1 uncertain)

r is the risk-free rate

X_t is price/payoff of the forward contract at t

Question: What is difference between X and F ?

Assumption: no frictions: costless storage, shorting, etc. (relax later)

USE LoOP

Two ways to get cocoa tomorrow: buy today (store) and buy forward

Buying today costs p_0 today

Buying forward costs $F_0(1)$ tomorrow or $\frac{F_0(1)}{1+r}$ today

USE LoOP

Two ways to get cocoa tomorrow: buy today (store) and buy forward

Buying today costs p_0 today

Buying forward costs $F_0(1)$ tomorrow or $\frac{F_0(1)}{1+r}$ today

$$\text{LoOP} \implies p_0 = \frac{F_0(1)}{1+r}$$

USE LoOP

Two ways to get cocoa tomorrow: buy today (store) and buy forward

Buying today costs p_0 today

Buying forward costs $F_0(1)$ tomorrow or $\frac{F_0(1)}{1+r}$ today

$$\text{LoOP} \implies p_0 = \frac{F_0(1)}{1+r} \text{ or } F_0(1) = (1+r)p_0$$

USE LoOP

Two ways to get cocoa tomorrow: buy today (store) and buy forward

Buying today costs p_0 today

Buying forward costs $F_0(1)$ tomorrow or $\frac{F_0(1)}{1+r}$ today

$$\text{LoOP} \implies p_0 = \frac{F_0(1)}{1+r} \text{ or } F_0(1) = (1+r)p_0$$

Generalizes to $T - t$ periods: $F_t(T) = (1+r)^{T-t}p_t$

ALTERNATIVE: REPLICATE THE FORWARD X

Forward payoff: $X_1 = p_1 - F_0(1)$

Replicating portfolio V : long one unit cocoa, short $(1 + r)p_t$ bonds

NB: Shorting $(1 + r)$ bonds $\equiv$ borrowing \$1

Portfolio costs $V_0 = 0$ today pays off $V_1 = p_1 - (1 + r)p_0$ tomorrow

So $X_0 - V_0 = 0$ (why?) and $X_1 - V_1 = F_0(1) - (1 + r)p_0$ (no risk)

No arb $\implies$ zero value at $t = 1 \implies F_0(1) = (1 + r)p_0$

FORWARD PRICES & EXPECTATIONS

FORWARD PRICES & EXPECTATIONS

Cocoa seller and chocolate maker want to hedge cocoa exposure

Both are bullish on cocoa

they think price will increase

Cocoa seller argues thus they should set a high forward price

Says forward price should be greater than $(1 + r)p_t$

What should the chocolate maker say?

PRACTICE EXERCISE: FORWARD PRICE OF A BOND

EXERCISE: FORWARD PRICE OF A BOND

The risk-free rate is 10%. Assume it does not change over time

What is the one-year forward price $F_0(1)$ of

- (i) a bond with current maturity one year
- (ii) a bond with current maturity two years

Relate the forward price to

- (a) the interest rate
- (b) the maturity of the bond
- (c) the expected spot price

FORWARD EXCHANGE RATES

Forward ex. rate: Forward price of one currency in another

Ex: 2y euro in USD in 8/23: $r_{\$} = 5.01\%$, $r_{\text{€}} = 2.98\%$, $\text{FX}_{\text{€} \rightarrow \$} = 1.087$

Claim to €1 at $t = 2$ has price $p_0 = \frac{\text{FX}_{\text{€} \rightarrow \$}}{(1 + r_{\text{€}})^2}$

Forward-price formula:

$$F_{\text{€} \rightarrow \$} = (1 + r_{\$})^2 p_0 = \left(\frac{1.0501}{1.0298} \right)^2 \times 1.087 \approx 1.130$$

RECAP

LECTURE 3: RECAP

Forwards useful for hedging but what is the forward price?

Found (absent storage costs) $F_0(T) = p_0(1 + r)$

Expectations and forward prices

Forward prices of bonds

Forward exchange rates

FIN